

CURRICULUM VITAE
Álvaro Ras Carmona, PhD

Name: **Álvaro** *Last Names:* **Ras Carmona**
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CONTACT INFORMATION

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Bioinformatician with 8 years of experience and a strong background in **artificial intelligence**, **data analysis**, and **biostatistics**. I have **advanced programming skills** and experience handling large volumes of biological data. My career has focused on the development and application of machine learning models—ranging from classical algorithms to advanced **deep learning** architectures and **transformers** (particularly encoded-only ones)—applied to biological problems. I have experience in the full model development cycle, including data preparation, training, and evaluation. I am currently participating in a drug repositioning project using artificial intelligence. I am the author of **16 scientific publications and book chapters** (9 as first or co-first author) and have made **14 presentations at national and international conferences**. I seek to contribute my technical and analytical expertise to the development of innovative solutions in the biotechnology and healthcare sectors.

STUDIES AND TITLES

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| Oct 2019 – Jul 2024 | PhD in Biomedical Research (Specialty: Bioinformatics)
Complutense University of Madrid (Spain)
<i>Thesis title:</i> Development of bioinformatics tools for the prediction of linear epitopes of B lymphocytes
<i>Grade:</i> Cum Laude |
| Oct 2018 – Sep 2019 | MSc Omics Data Analysis (Bioinformatics)
Universidad de Vic - Universidad Central de Cataluña (Spain)
<i>Average grade:</i> 8,1 |
| Sep 2014 – Jul 2018 | Degree in Biotechnology
Universidad Francisco de Vitoria (Spain)
<i>Average grade:</i> 7,8 |

OTHER OFFICIAL UNIVERSITY DEGREES

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| Sep 2014 – Jul 2018 | Expert Degree in Biotechnology Research
Universidad Francisco de Vitoria (Spain)
<i>Average grade:</i> 8,3 |
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COURSES

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| Sep 2026 | 5th AIHUB Summer School
Attendance at the AIHUB Summer School for Early-Career Researchers in AI Applied to Science
<i>Duración:</i> 1 week |
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May 2026	Workshop on Parallel Supercomputing with GPUs on HCP DRAGO Course organized by the Deputy Secretary-General for Human Resources in collaboration with the Deputy Secretary-General for Information Technology. <i>Duration:</i> 20 hours
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RESEARCH EXPERIENCE

Jun 2025 – Present	Postdoctoral Researcher Macromolecular Assembly Structure Group, “Margarita Salas” Biological Research Center, Spanish National Research Council (CSIC), Madrid <i>Topic:</i> Implementation of predictive solutions using pre-trained natural language processing models applied to proteins (encoder-only) and deep learning models, as well as the development of innovative algorithms in bioinformatics (artificial intelligence, drug discovery, and biochemistry)
Jul 2024 – Jun 2025	Post-Doctoral Research Scientist Immunomedicine group, Department of Immunology, School of Medicine, Complutense University of Madrid (Spain) <i>Topic:</i> Study of the sequence space of linear B-cell epitopes through the use of dimensionality reduction algorithms (t-SNE and UMAP) (Bioinformatics and Immunology).
Oct 2019 – Jul 2024	Pre-Doctoral Research Scientist Immunomedicine group, Department of Immunology, School of Medicine, Complutense University of Madrid (Spain) <i>Topic:</i> Linear B cell epitope prediction with supervised machine learning models (Bioinformatics and Immunology).
May 2019 – Sep 2019	Master student: Completion of the MSc final project Immunotherapy of cancer research group, advanced therapies area, Universidad Francisco de Vitoria (Spain). <i>Topic:</i> Bioinformatic analysis of the methylome and differentially methylated genes in colorectal cancer subtypes (Metagenomics and Biostatistics)
Jan 2018 – Jul 2018	Degree student: Completion of the degree final project Centro de Investigaciones Biológicas Margarita Salas, Consejo superior de Investigaciones científicas (CSIC), Madrid (Spain) <i>Topic:</i> Establishment and optimization of the TFIIF purification process in <i>Saccharomyces cerevisiae</i>. (Structural biology)

INTERNATIONAL INTERSHIP

<i>Research group</i>	Computational Modeling in Systems Biomedicine
<i>Place</i>	Dipartimento di Scienze del Farmaco e della Salute, University of Catania, Catania, Italy
<i>Supervisor</i>	Prof. Francesco Pappalardo
<i>Dates</i>	19 September – 1 December
<i>Financing (Grant)</i>	Internship financed by the European Federation of Immunological Societies and Immunology Letters

<i>Topic</i>	Prediction of the next variant of concern of the SARS-CoV-2 virus by the application of a genetic algorithm implemented de novo in Python.
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PARTICIPATION IN RESEARCH PROJECTS

2023 – 2026	<i>“Development of an epitope vaccine against rhinovirus”</i> . EPI4RV (PID2022-136662OB-I00). Member of the research group.
2020 – 2022	<i>“Anticipation and prevention of covid-19 in the Madrid Region”</i> . ANTICIPA-CM (PR38/21-33). Member of the research group.
2020 – 2022	<i>“Cross-immunity of bacterial formulations against viruses that cause recurrent respiratory diseases”</i> . INMUNOTEK, S.L. (337-2020). Member of the research group.
2020 – 2022	<i>“Verification of the accuracy of existing in silico T cell epitope prediction algorithms vs. the experimental data that CTL generates”</i> . CELLULAR TECHNOLOGY LIMITED (282-2020). REF: 4157895. Member of the research group.
2020 – 2021	<i>“e-FACS: a platform for virtual realization of flow cytometry”</i> . Innova-Docencia projects 2020-21. Complutense University of Madrid. REF: 243. Member of the research group.

SKILLS

- **Machine Learning & Deep Learning:** Python · R · PyTorch · TensorFlow · Scikit-learn · Deep Learning · Transformers · Mixture of Experts (MoE) · MLPs · Large Language Models (LLM) · RAG · NLP · Protein Language Models (ESM-2, ProtBERT, ProtT5 ...) · Genetic Algorithms · Binary Classification · Cross-Validation y Model Evaluation · Hyperparameter Optimization with Optuna
- **Biología Computacional:** BioPython · Biostatistics · Protein Structure Prediction · AlphaFold3 · In Silico Peptide Design · BindCraft · BinderFlow · Protein–Protein Interaction (PPI) Prediction · AlphaPulldown · Computational chemistry · Scipion-Chem · Immunoinformatics
- **Análisis de Datos & Visualización:** NumPy · Pandas · SQL · UMAP · t-SNE
- **Computación Científica & Infraestructura:** HPC · GPU Computing · Slurm · Docker · Ollama · Linux · Bash · Git/GitHub

LANGUAGES

Spanish	Native
English	B2: Official title issued by the University of Cambridge: First Certificate Exam (FCE).

GRANTS AND CONTRACTS AWARDED

2024	Modification of the pre-doctoral contract for research personnel in training (CT82/20-CT83/20):
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- *Amendment in question:* "Having read and approved the doctoral thesis on 08.07.2024, [...], pre-doctoral contracts shall include in their object a period of post-doctoral orientation for a maximum period of twelve months. It is therefore modified on the occasion of obtaining the degree of doctor".
 - *Contract end:* 08/07/2025
- 2022 **Award of short stay scholarship:** *EFIS-Immunology Letters Short-Term Fellowship*
- *Funding agency:* European Federation of Immunological Societies and Immunology Letters
 - *Period covered by the scholarship:* 19/09/2022 – 1/12/2022
- 2022 **Award of pre-doctoral contract** for research personnel in training. Call 2020 (Código de la plaza: *CT82/20-CT83/20*)
- *Contract duration:* 01/06/2022 – 31/05/2026
- 2021 **Awarding of a position as** “Personal Investigador de Proyectos (PAI-I). UCM” **(Contract)** (Código de la plaza: *PAII87/20-15/2020-27*).
- *Contract duration:* 15/01/2021 – 31/12/2021
 - *Extended up to:* 31/05/2022
- 2014 - 2017 Award during the corresponding academic years of the **academic excellence scholarships** granted by the Universidad Francisco de Vitoria in honor of the grades obtained.

ACCADEMIC/TEACHING EXPERIENCE

- 2025 – 2026 **Advising a master's student.** Master's in Bioinformatics and Computational Biology. Autonomous University of Madrid (UAM)
- **Master's Thesis Title:** Development of AI Tools for Drug Discovery Against Sleeping Sickness
 - **Master's Thesis Grade:** 8.5
- 2022 – 2023 **Teaching in Medicine degree.** Complutense University of Madrid (UCM), Faculty of Medicine. Subject *Immunology*. 12 hours.
- **Positive evaluation in Docentia UCM** (Average grade: 9,49).
- 2021 – 2022 **Teaching in Medicine degree.** Complutense University of Madrid (UCM), Faculty of Medicine. Subject *Immunology*. 24 hours.
- **Positive evaluation in Docentia UCM** (Average grade: 8,67).
- 2020 – 2021 **Teaching in Medicine degree.** Complutense University of Madrid (UCM), Faculty of Medicine. Subject *Immunology*. 15 hours.

PUBLICATIONS

Articles

- Di Salvatore, V.*, Maleki, A.*, Mohajer, B., **Ras-Carmona, A.**, Russo, G., Reche, P. A., & Pappalardo, F. 2025. Exploring SARS-CoV-2 spike protein mutations through genetic algorithm-driven structural modeling. *Bioinformatics advances*. <https://doi.org/10.1093/bioadv/vbaf288>. *Equal contribution

- **Ras-Carmona, A.**, Lehmann, A. A., Reche, PA. 2025. Similarity to Self-Antigens Shapes Epitope Recognition from Viruses Under Autoimmune and Infectious Disease. *International Journal of Molecular Sciences*. <https://doi.org/10.3390/ijms26136041>
- **Ras-Carmona, A.**, Lafuente, E. M., Reche, PA. 2025. Analysis of gender and scientific output of researchers in the Spanish University. *Journal of Education Culture and Society*. <https://doi.org/10.15503/jecs2025.2.859.871>
- Pelaez-Prestel, H.F., Gonzalez-Martin, F., **Ras-Carmona, A.**, Rocha, A., Cabañas, C., Lafuente, E. M., Reche, P.A. 2024. Oral squamous cell carcinomas drive monocytes into immunosuppressive CD25+CD163+CD206+ macrophages. *Oral Oncology*. <https://doi.org/10.1016/j.oraloncology.2024.107078>.
- Fernandez, S.A.*, Pelaez-Prestel, H.F.*, **Ras-Carmona, A.**, Mozas-Gutierrez, J., Reyes-Manzanas, R., Reche, P.A. 2024. Eucalyptus Essential Oil Inhibits Cell Infection by SARS-CoV-2 Spike Pseudotyped Lentivirus. *Biomedicines*. <https://doi.org/10.3390/biomedicines12081885>. *Equal contribution
- **Ras-Carmona, A.**, Reche, PA. 2024. Analysis of Virus-Specific B Cell Epitopes Reveals Extensive Antigen Degradation Prior to Recognition. *Cells*. <https://doi.org/10.3390/cells1313107>
- Bodas-Pinedo, A.* , Lafuente, E. M.* , Pelaez-Prestel, H. F., **Ras-Carmona, A.**, Subiza, J. L., & Reche, P. A. 2023. Combining different bacteria in vaccine formulations enhances the chance for antiviral cross-reactive immunity: a detailed in silico analysis for influenza A virus. *Frontiers in immunology*. <https://doi.org/10.3389/fimmu.2023.1235053>. *Equal contribution
- **Ras-Carmona, A.**, Lehmann, A. A., Lehmann, P. V., & Reche, P. A. 2022. Prediction of B cell epitopes in proteins using a novel sequence similarity-based method. *Scientific reports*. <https://doi.org/10.1038/s41598-022-18021-1>
- Ballesteros-Sanabria L*, Pelaez-Prestel H.F*, **Ras-Carmona A***, Reche, PA. 2022. Resilience of Spike-Specific Immunity Induced by COVID-19 Vaccines against SARS-CoV-2 Variants. *Biomedicines*. <https://doi.org/10.3390/biomedicines10050996>. ***Equal contribution**
- **Ras-Carmona A**, Pelaez-Prestel H.F, Lafuente E.M, Reche, P.A. 2021. BCEPS: A Web Server to Predict Linear B Cell Epitopes with Enhanced Immunogenicity and Cross-Reactivity. *Cells*. <https://doi.org/10.3390/cells10102744>
- **Ras-Carmona A**, Gomez-Perosanz M, Reche PA. 2021. Prediction of unconventional protein secretion by exosomes. *BMC Bioinformatics*. https://doi.org/10.1007/978-1-0716-0389-5_23.
- Gomez-Perosanz M*, **Ras-Carmona A***, Lafuente EM, Reche PA. 2020. Identification of CD8+ T cell epitopes through proteasome cleavage site predictions. *BMC Bioinformatics*. <https://doi.org/10.1186/s12859-020-03782-1>. ***Equal contribution**

Book chapters

- Gomez-Perosanz M, **Ras-Carmona A**, Reche PA. 2020. *Immunoinformatics*. In: Namrata Tomar (ed.). *Methods in Molecular Biology*, vol. 2131. https://doi.org/10.1007/978-1-0716-0389-5_23.

Workshop papers

- Workshop paper in “2023 *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*”. Avisa Maleki, **Alvaro Ras-Carmona**, Elena Crispino, Valentina Di Salvatore, Giulia Russo, Pedro Antonio Reche and Francesco Pappalardo. Genetic Algorithm-Based Prediction of Emerging SARS-CoV-2 Variants: A Computational Biology Perspective. Istanbul, Türkiye.
- Workshop paper in “2022 *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*”. Avisa Maleki*, **Alvaro Ras-Carmona***, Valentina Di Salvatore, Giulia Russo, Elena Crispino and Francesco Pappalardo. Genetic algorithm application

for the prediction of potential SARS-CoV-2 new variant of concern. Las Vegas NV, USA. ***Equal contribution**

- Workshop paper in “2019 *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*”. Gomez-Perosanz M, **Ras-Carmona A**, and Reche PA. Prediction of proteasomal cleavage sites using PCPS. San Diego CA, USA.

CONGRESSES

- 2025 **Participated as a speaker** in the “*Corner Life Experience*” session held during the “*4th St. Albert the Great Congress*”
- 2024 **Poster** in the “*III Congreso San Alberto Magno*”. **Álvaro Ras-Carmona** and Pedro A. Reche. El análisis de epítomos virales de linfocitos B revela la necesidad de un amplio procesamiento del antígeno para su reconocimiento.
- 2024 **Participation as a speaker** at events held during the “*III Congreso San Alberto Magno*”
- 2024 **Poster** in the “*XII Jornadas Científicas de la Sociedad de Inmunología de la Comunidad de Madrid*”. **Álvaro Ras-Carmona**, Pedro A. Reche. El análisis de epítomos virales de linfocitos B revela la necesidad de un amplio procesamiento del antígeno para su reconocimiento.
- 2023 **Participation as a speaker** at events held during the “*II Congreso San Alberto Magno: Afrontando Retos Desde La Ciencia*”
- 2023 **Poster** in the “*II Congreso San Alberto Magno: Afrontando Retos Desde La Ciencia*”. **Álvaro Ras-Carmona**, Hector F. Pelaez-Prestel, Esther M. Lafuente and Pedro A. Reche. Inteligencia artificial e inmunología: Desarrollo de un modelo de aprendizaje automático para la predicción de epítomos lineales de linfocitos B
- 2023 **Poster** in the “*XI Jornadas Científicas de la Sociedad de Inmunología de la Comunidad de Madrid*”. **Álvaro Ras-Carmona**, Alexander A. Lehmann, Paul V. Lehmann, Pedro A. Reche. BepiBlast: Un servidor web para predecir epítomos lineales mediante un nuevo método basado en similitud de secuencias.
- 2023 **Oral communication** in the “*44 Congreso de la sociedad de la Sociedad Española de Inmunología*”. **Álvaro Ras-Carmona**, Alexander A. Lehmann, Paul V. Lehmann, Pedro A. Reche. Desarrollo de un nuevo método para la predicción de epítomos lineales de linfocitos B basado en la similitud de secuencias.
- 2022 **Poster** in the “*X Jornadas Científicas de la Sociedad de Inmunología de la Comunidad de Madrid*”. **Álvaro Ras-Carmona**, Hector F. Pelaez-Prestel, Esther M. Lafuente and Pedro A. Reche. BCEPS: Hacia una predicción de epítomos lineales de linfocitos B con un alto grado de inmunogenicidad y reactividad cruzada
- 2022 **Poster** in the “*X Jornadas Científicas de la Sociedad de Inmunología de la Comunidad de Madrid*”. Laura Ballesteros-Sanabria, Hector F. Pelaez-Prestel, **Álvaro Ras-Carmona**, Pedro A. Reche. Resilience of Spike-Specific Immunity Induced by COVID-19 Vaccines Against SARS-CoV-2 Variants.
- 2021 **Poster** in the “*5ª Convocatoria EDUCM PhDay Complutense*”. **Álvaro Ras-Carmona**, Hector F. Pelaez-Prestel, Esther M. Lafuente and Pedro A. Reche. BCEPS: Hacia una predicción de epítomos lineales de linfocitos B con un alto grado de inmunogenicidad y reactividad cruzada. **Award-winning poster**
- 2021 **Poster** in the “*6th European Congress of Immunology*”. **Álvaro Ras-Carmona**, Marta Gomez-Perosanz, and Pedro A. Reche. ExoPred: The first method for predicting vertebrata secreted proteins via exosome using random forest algorithm.
- 2021 **Poster** in the “*42 Congreso de la sociedad de la Sociedad Española de Inmunología*”. **Álvaro Ras-Carmona**, Marta Gomez-Perosanz, and Pedro A. Reche. ExoPred: A web-based tool for predicting proteins secreted by exosomes using Random Forests.
- 2020 **Poster** in the “*4ª Convocatoria EDUCM PhDay Complutense*”. **Álvaro Ras-Carmona**, Marta Gomez-Perosanz, and Pedro A. Reche. Prediction of vertebrata secreted proteins via exosome by the application of random forest algorithm.